

Model cards, mathematical purpose and justified exclusions

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MODEL-1 / Demand elasticity / AIDS

Status: Not estimated. Data: Household price/expenditure/category shares. Method: Deaton-Muellbauer demand system; adding-up, homogeneity and symmetry. Limitation: No harmonized household/category prices and expenditure panel. GDP and food production cannot substitute. Acquire surveys, test restrictions and hold out regions.. Decision changed if supported: Product mix, affordability, pack and price choices.

MODEL-2 / Trade gravity / PPML

Status: Not estimated. Data: Bilateral HS quantities/values, tariffs and structural access. Method: Santos Silva-Tenreyro PPML; origin/destination-time and pair effects as identified. Limitation: Single Nigeria HS1001 preview has missing net weight; broad WDI imports do not identify attainable capture. Acquire bilateral product flows and access controls.. Decision changed if supported: Origin/corridor entry and trade counterfactual.

MODEL-3 / Supply benchmark / OLS

Status: Executed and rejected for investment use. Data: Annual food-production index log growth. Method: Lagged growth pooled OLS, country CR1 clustering. Limitation: Training 2002-2018; retrospective 2019-2022 holdout. Tiny gain, target mismatch, instability; no causal or 2051 forecast.. Decision changed if supported: Retain benchmark and prioritize actual product/operating evidence.

MODEL-4 / FX/commodity volatility

Status: Not estimated. Data: Aligned procurement currency, commodity, basis and hedge series. Method: Engle ARCH/GARCH; Granger cointegration/error correction where suitable. Limitation: No matching exposure series or hedge basis. Tests must establish stationarity, breaks and forecast value.. Decision changed if supported: Hedge, collateral, pass-through and liquidity design.

MODEL-5 / Crop/climate impact

Status: Not estimated. Data: Geocoded crops, calendars, hazards, water and adaptation. Method: Crop-process model or identified yield/weather response; IPCC informs mechanisms. Limitation: Country water stress and global climate assessment cannot price actual assets. Obtain basin/asset hazards and adaptation costs.. Decision changed if supported: Origin, backup, adaptation or no-go decisions.

MODEL-6 / Marketing / causal lift

Status: Designed; no observed Olam effect. Data: Randomized or credible matched route/customer data. Method: Randomized cluster ITT; modern DID for nonrandom timing with pre-trends/overlap. Limitation: Sample/power need baseline variance and ICC. Record noncompliance, spillover, cannibalisation, attrition and sustained paid repeat.. Decision changed if supported: Spend, service, price/pack and scaling decisions.

MODEL-7 / Customer repeat

Status: Not estimated. Data: Linked customer orders, promotion and actual collections. Method: Cohort/survival or hierarchical model, validated by customer/time holdout. Limitation: No linked customer panel; repeated shipments may be loading rather than paid adoption.. Decision changed if supported: Retention/credit limits and service.

MODEL-8 / DCF / cash

Status: Executed provisional template. Data: Chosen normalized physical volume, price, cost, capex and credit inputs. Method: FCFF identity; finite discounted sum; investment order root. Limitation: 18 matched mode/world cases, actual taxes/FX/monthly cash absent; 195 country cases remain threshold-only.. Decision changed if supported: Wait/partner/asset and economic thresholds.

MODEL-9 / Robust/adaptive allocation

Status: Executed teaching cases only. Data: Feasible actions, state payoffs, budgets and learning. Method: Minimax regret; CVaR; Bayesian EVSI; constrained network LP. Limitation: Chosen payoffs, priors and dependencies. No empirically optimal all-country allocation or mathematical novelty.. Decision changed if supported: Capped experiments, staged commitments and network constraints.

Solow / Romer

Growth and knowledge constraints. Scenario framing only; no company-sales coefficients or convergence forecast. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.

Nordhaus

Climate/economy integration. Conditional scenario architecture; no aggregate damage copied into a basin. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.

Deaton / Muellbauer

Consumption, substitution and welfare. Demand module data requirements; not fitted. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.

Engle / Granger

Volatility and cointegration. Proposed exposure time-series module; not fitted. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.

Krugman

Scale, transport and trade location. Location mechanism only; firms/access still require evidence. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.

Leontief

Input-output dependencies. Physical/material balance discipline; no fitted national IO multipliers. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.

Markowitz / Sharpe

Dependencies and systematic risk. Risk/cost-of-capital concepts; chosen stress is not fitted covariance or WACC. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.

Ostrom

Shared water/supplier governance. Local rights and institutional design gate; not portable coefficient. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.

Card / Angrist / Imbens; Callaway / Sant'Anna

Causal inference. Pilot identification requirements; no realized treatment estimate. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.

Institutions and development research

Execution and institutional paths. Qualitative feasibility/uncertainty gate; no deterministic country score. Attribution is to original contributors, not a claim that all coauthors won the same prize; official prize context in NOBEL_2003/2018/2021. Other foundations are background and unimplemented here.